(a) Rheumatism
(b) Nervous shock
(c) Muscular dystrophy
(d) Cerebral hemorrhage

Ans. (c) : X-linked recessive diseases most often occur in males. Males have only one X chromosome. A single recessive gene on that X chromosome will cause the disease. The Y chromosome is the other half of the XY gene pair in the male. However, the Y chromosome dosen't contain most of the genes of the X-chromosome. Because of that, it doesn't protect the male. Diseases such as hemophilia and Duchenne Muscular dystrophy occur from a recessive gene on the X-chromosome.

104.Which one of the following vitamins is considered to be a hormone ?
(a) A (b) B
(c) C (d) D

Ans. (d) : The body makes vitamin D in a chemical reaction that occurs when sunlight hits skin. This reaction produces cholecalciferol, and the liver converts it to calcidiol. The kidneys then convert the substance to calcitriol, which is the active form of the hormone in the body.
Vitamin D helps the body absorb calcium so that blood calcium levels are at the ideal point. This helps enable the mineralization of bone that is required for strong, healthy bones. Yet this is just one function of the hormone.

105.Rennin and lactase, the enzymes required to digest milk, disappear in the human body by the age of
(a) two (b) three
(c) five (d) eight

Ans. (d) : By the age of eight years, renin and lactase disappear in the human body. Rennin is a proteolytic enzyme found in the gut of many animals and synthesized by the chief cells located in the mucosal epithelium of the stomach. Its function is to coagulate the proteins of milk and aid in its digestion.
Lactase is an enzyme that is required by the body for cleaving lactose into glucose and galactose. Lactose is a sugar that is found in milk or its products. This enzyme is only found in the small intestine of mammals.

106. Haemoglobin is dissolved in the plasma of
(a) frog (b) fish
(c) man (d) earthworm

Ans. (d) : Just like other animals, earthworms need to transport blood around the body to survive.
Earthworm blood-unlike that of most invertebrates contains haemoglobin, a protein in red blood cells to carry oxygen around their bodies.

107.Phenylketonuria is an example of an in born error of metabolism. This "error" refers to
(a) hormonal overproduction
(b) atrophy of endocrine glands
(c) inherited lack of an enzyme
(d) non disjunction

Ans. (c) : Phenylketonuria is an example of an born error of metabolism. This "error" refers to inherited lack of an enzyme.
Mutations in the PAH gene cause phenylketonuria.
The PAH gene provides instructions for making an enzyme called phenylalanine hydroxylase. This enzyme converts the amino acid phenylalanine to other important compounds in the body.

108.Assertion (A): A lock of Einstein's hair, if scientists could locate it and extract its DNA, could help in producing another Einstein, by cloning.
Reason (R): The DNA extracted from the cell of an embryo at an early stage of development, can be transferred to individual eggs which in turn can be implanted into the uterus of a surrogate mother to give birth to an identical off spring.
In the context of the above two statements, which one of the following is correct ?
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not a correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Ans. (a) : A lock of Einstein's hair, if scientists could locate it and extract its DNA, could help in producing another Einstein, by cloning. Because the DNA extracted from the cell of an embryo at an early stage of development, can be transferred to individual eggs which in turn can be implanted into the uterus of a surrogate mother to give birth to an identical off spring.

ENVIRONMENT AND ECOLOGY

109.Farm land tends to erode most rapidly when planted with
(a) sorghum (b) potato
(c) wheat (d) clover

Ans. (a) : Farm land tends to erode most rapidly when planted with sorghum. The transition to agriculture from natural vegetation often can not hold on to the soil and many of these plants, such as coffee, cotton, palm oil, soyabean and wheat, can actually increase soil erosion beyond the soil's ability to maintain it self. The effects of soil erosion go beyond the loss of fertile land. It has led to increased pollution and sedimentation in streams and rivers, clogging these water ways and causing declines in fish and other species.

110.Match List I with List II and select the correct answer from the codes given below the lists :

List I (Crops)	List II (Crop Pests)
I. Rice	(A) Aphid
II. Wheat	(B) Gundhi bug
III. Sugarcane	(C) Greasy cutworm
IV. Gram	(D) Top shoot borer moth
	(E) Bollworm

Codes:
(a) I-B II-C III-D IV-E
(b) I-C II-A III-B IV-D
(c) I-B II-A III-D IV-C
(d) I-E II-D III-A IV-C

Ans. (c) : The damage caused by agricultural pests is one that is devastating. On a global scale, the losses caused by pests in arable crops accounts for about 10% - 15% of the total yield value.
Aphids are 1 – 3 mm, soft bodies insects that can be green, grey, or black. It's found in wheat. Gundhi bug found in rice, in the case of heavy infestation, the whole year ahead may become devoid of mature grains.
Buggy odour in rice field during milky stage. Greasy cutworm's young larva feed on the epidermis of the leaves.
Older larva come out at night and feed young plants by cutting their stems.

111.Which of the following is/are used as biofertilizers ?
I. Azolla
II. Bluegreen algae
III. Alfalfa.
Select the correct answer using the codes given below.
Codes:
(a) II alone (b) I and II
(c) I and III (d) I, II and III

Ans. (b) : 'Biofertilizer' is a substance which contains living microorganism which, when applied to seed, plant surfaces, or soil, colonizes the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant.
Biofertilizers add nutrients through the natural process of fixing atmospheric nitrogen, solubilizing phosphorus and stimulating plant growth through the synthesis of growth promoting substances. There are may types of bioferlizers- Azotobacter, clostridium Anabaena, Nostoc, Rhizobium, Frankia, etc.

112. Which one of the following crops enriches the nitrogen content in soil ?
(a) Potato (b) Sorghum
(c) Sunflower (d) Pea

Ans. (d) :
• Pea crops enrich the soil with nitrogen.
• Pea and beans are nitrogen-fixing crops and hence they are called Legumes.
• They fix nitrogen in the soil with the help of nitrogen-fixing bacteria present in their roots called Rhizobium.
•Nitrogen is essential for plant for its photosynthesis as it is a main component in chlorophyll and for plant rapid growth.

113. If water pollution continues at its present rate, it will eventually
(a) stop water cycle
(b) prevent precipitation
(c) make oxygen molecules unavailable to water plants
(d) make nitrate unavailable to water plants

Ans. (c) : If water pollution continues at its present rate, it will eventually make oxygen molecules unavailable to water plants.
Plants require specific nutrients, which they get from the soil, water and air. Polluted water might kill some of those nutrients, further denying the plant the ability to get them.

CURRENT AFFAIRS

114.Which of the following are the purposes to which the National Renewal Fund has been set up ?
I. To restructure sick small scale industries.
II. To help workers likely to be displaced due to retrenchment in the process of industrial restructuring.
III. To modernise existing industrial units.
IV. To help refugees from Bangladesh, Sri Lanka etc.
Choose the correct answer from the code given below :
Codes:
(a) I and II (b) I and III
(c) I and IV (d) II and IV

Ans. (a) : The National Renewal Fund (NRF) was set up in February, 1992 by a Government resolution which provided for its operation for a limited period of time up to a maximum of ten years. Assistance from NRF has been provided for the Voluntary Retirement Scheme (VRS) and the scheme of counseling retraining etc. of workers rationalised from the organised sector.